



Ajit Pai investigated for corruption

FCC Chairman Ajit Pai is being investigated by his own corporation for potential corruption. <http://digit.in/paino>



World's largest SSD

Samsung has unveiled the world's largest solid state drive with a whopping 30 TB of storage - enough for 5,700 FHD movies. <http://digit.in/ssungSSD>

- (ii) different organs and tissues in a single organism are heterogeneous
- (iii) different cells in the same organ are also heterogeneous.

Translation: Each cell in your body is like an individual unit with its own RNA, DNA and protein conversions. While there is ongoing research being done to estimate the number of types of cells in a human body, the best explanations that have been provided until date have a staggering range – from millions to even trillions.

BEEN AROUND FOR AGES

Speaking of intrigued, did you know that heterogeneity at a cellular level is not really a new finding? It has been mentioned in scientific texts that date back to the late 1950s! In recent times though, single cell sequencing (SCS) has come into the limelight due to its applications to the field of cancer research. Accurate determination of SCS will help understand the lineage of the cell in question as well as help detect any residual disease. SCS could also help our basic understanding of mammalian development, immunity systems or even the central nervous system. Any new findings about these complex systems are of paramount importance to the world of biology.

THE PROCESS

There has been a significant improvement in the process of DNA sequencing. To fully and correctly sequence a single cell's DNA, there are three techniques that need to be applied -

- (i) Isolate the single cell
- (ii) Augment or expand the genome. Doing so accurately as well as efficiently is of utmost importance here.
- (iii) Sequence the DNA of the single cell

By correctly applying these techniques, you can compare the differences between the genomes of cells in say, the same organ or tissue.

APPLICATIONS

There has been a lot of talk in the world of healthcare as well as other-

wise about the various applications of single cell DNA sequencing. Here's an overview of a few.

Cancer Research

In the past, doctors and researchers tested with sampled bulk tissues. The problem here was that these tissues contained multiple cells with a myriad of messages that the researchers were then trying to (in most cases incorrectly) decode. But all cells within the



Scientists have been trying to find the cure for cancer since the invention of the microscope. SCS may have the answer

tissue culture are different, remember? With the advent of SCS, we could now better study intratumor heterogeneity, rates of mutation, rare tumor cells etc.

The composition of cancer involves multiple clones of the affected cell. Especially in advanced tumors, there will be an even larger number of clones, many of which will be unique in nature given their heterogeneity. Each of these unique subclones will then have their own set of mutations. This effectively means that all of these cells will react differently to the same therapy. Then there are other complexities like a subclone that might only make up for 0.5% of a primary tumor could then turn into the dominant clone if the patient were to experience a relapse. This is what makes molecular profiling even more important. Single Cell DNA sequencing has the potential to single-handedly change the way we treat cancer across the globe!

Stem Cell Research:

One of the biggest challenges in the world of stem cell research is identifying a bonafide stem cell (SC). Stem cell research uses a lot of data acquired from single cells. Everything from its molecular structure to its cellular behaviour become central to stem cell research. Given the recent advancements in the field of single cell DNA sequencing, it is now possible to study the high throughput from single cells.

There are also methods found to continuously observe and study the cellular behaviour of single cells in a non-invasive manner.

All of this translates to a huge success for the world of medicine and health care solutions. Researchers now have more accurate means to studying everything from human growth, cell redevelopment, cures to diseases like Parkinsons or Huntington's, therapies for spinal cord injuries and so much more. This is just the tip of the iceberg.

Any more detail, and we will be lost in a world of jargon.

CHALLENGES

Every silver cloud has a gray lining too. This beautiful world of single cell sequencing has its own challenges. For SCS studies to be statistically significant, a very large number of single cells need to be isolated and studied. This number could vary from hundreds to thousands, for it to be relevant to the study in question. This would mean mountain loads of data and the need to sift through and study this data in an efficient manner without losing the individuality of every single cell.

But the challenges that pave the path to success make the work accomplished by single cell DNA sequencing an absolute success. There is groundbreaking research that is being done in the field every passing day, and a cure for cancer might actually be possible! 